

Mortality-related meaning enhances predicted memorability in paintings beyond hierarchical visual features

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Significant: Why some images are remembered better than others is often treated as a property of visual structure that computational models can predict with high accuracy. Yet an important question remains: do such models fully explain memorability, or does affective meaning contribute beyond what current vision-based models capture? We used mortality-related paintings as a test case because they are visually diverse yet emotionally salient. We show that their memorability advantage is partly explained by emotional arousal and is less fully accounted for by hierarchical visual features than the memorability of non-death paintings. Current memorability models therefore capture substantial, but incomplete, structure in affectively charged images. These findings connect stimulus-driven accounts of memorability with emotional memory research and suggest that future models should incorporate affective relevance alongside visual structure.

Abstract:

Memorability is often described as a stable, stimulus-driven property of images that can be predicted from visual features by computational models trained on large-scale human memory data. Yet it remains unclear whether such models fully explain memorability, or whether psychologically salient meaning contributes beyond hierarchical visual features. We addressed this question using mortality-related paintings, which are visually heterogeneous but emotionally salient. Across 4,345 paintings from the Art Institute of Chicago, we combined predicted memorability (ResMem), hierarchical visual features from seven AlexNet layers, and affective content estimates.

Mortality-related paintings showed higher predicted memorability than non-death paintings ($d = 0.22$, $p < .001$). Mortality-related and non-death paintings were also separable throughout the AlexNet hierarchy, with the strongest separation in later network layers, indicating that hierarchical visual features capture substantial structure in mortality-related art. However, these same features predicted memorability less fully for death paintings ($r = 0.62$) than for non-death paintings ($r = 0.82$) at the highest layer, indicating a systematic prediction gap. Mediation analysis further showed that emotional arousal statistically accounted for the memorability advantage of mortality-related paintings (indirect effect = 0.021, 95% CI [0.013, 0.028]), with the direct effect reduced to non-significance after arousal was included. This result identifies arousal as a plausible source of the unexplained variance. Mortality-related subcategories further showed heterogeneous layer profiles, consistent with multiple visual-semantic routes rather than a single shared representation.

These findings suggest that predicted memorability reflects both identifiable visual-semantic structure and additional affective salience. More broadly, they indicate that current vision-based memorability models capture substantial, but incomplete, structure in affectively charged images.

Introduction

Why are some images remembered better than others? Research on visual memorability has shown that some images are consistently remembered better than others across observers, suggesting that memorability reflects stable properties of the stimulus rather than noise alone. This work has led to the view that memorability is a reliable, stimulus-driven image property that can be studied systematically across diverse visual materials (Isola et al., 2014; Khosla et al., 2015). Under this view, memorability provides a useful window into how image properties shape what is retained in memory. It also raises the possibility that these stable properties can be characterized and predicted computationally.

This stimulus-driven view of memorability has motivated growing efforts to model it computationally. A large body of work has shown that image memorability can be predicted from visual features using machine-learning and deep-learning approaches, including models trained directly on large-scale human memory data (Khosla et al., 2015; Bainbridge, 2020). This computational framework has been influential because it suggests that memorability is not merely a byproduct of observer variability but is at least partly embedded in the image itself. Recent work further indicates that this logic extends beyond photographs to artworks, whose memorability also shows systematic regularities and can be predicted at meaningful levels (Davis & Bainbridge, 2023). At the same time, this progress raises a deeper question: what kinds of information do current memorability models actually capture?

Despite the success of computational memorability models, a fundamental question remains. Although current models can predict memorability from visual features, it is still unclear whether visual structure alone fully accounts for why some images are remembered better than others. Images differ not only in visual structure but also in semantic meaning and affective salience, both of which are known to shape memory (LaBar & Cabeza, 2006; McGaugh, 2015). Such higher-order properties may contribute to memorability in ways that hierarchical visual features capture only indirectly, even when they are partially correlated with those features. This leaves an important interpretive question: Do current models capture memorability itself or the subset of memorability that is most visible in image structure?

Mortality-related paintings provide a particularly useful test case for this question. On the one hand, mortality-related content is visually heterogeneous: it can appear in symbolic forms, religious scenes, portraits, landscapes, depictions of violence, or memorial contexts, rather than as a single uniform visual category. On the other hand, death-related imagery is affectively and conceptually salient, making it a plausible case in which memorability may depend not only on visual structure but also on higher-order meaning. This logic is consistent with evidence that emotionally salient information is preferentially retained in memory and with broader theoretical work suggesting that mortality-related content is a distinctive source of psychological salience (LaBar & Cabeza 2006). If memorability were fully reducible to hierarchical visual structure, then any memorability advantage associated with mortality-related paintings should be largely captured by vision-based features alone. If not, mortality-related art should reveal a gap between what current visual models explain and what makes such images memorable.

To address these questions, we analyzed 4,345 paintings from the Art Institute of Chicago (488 mortality-related; 3,857 non-death). We combined ResMem-predicted memorability,

hierarchical visual features from seven AlexNet layers, and CLIP-derived affective content estimates. We tested (i) whether death-themed paintings show a memorability advantage, (ii) whether hierarchical visual features capture this advantage, (iii) whether emotional arousal helps explain variance that visual features do not, and (iv) whether mortality-related subcategories form a single representational class or heterogeneous profiles. Used this way, mortality-related art becomes a critical test case for whether memorability is fully explained by hierarchical visual structure or also shaped by affective and conceptual salience.

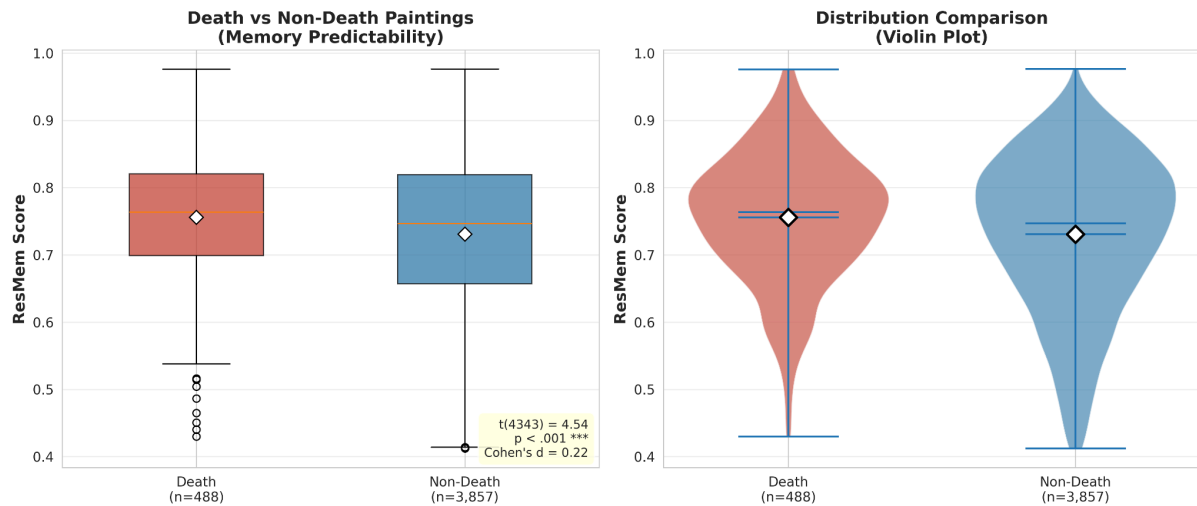
Results

Section 1. Mortality-related paintings show higher predicted memorability

We first tested whether mortality-related paintings differed from other paintings in predicted memorability. Across 4,345 paintings from the Art Institute of Chicago, 488 were classified as mortality-related and 3,857 as non-death. Memorability was estimated with ResMem (1), a deep convolutional model trained on large-scale behavioral memory data.

Mortality-related paintings showed higher predicted memorability than non-death paintings ($M = 0.756$, $SD = 0.096$ vs. $M = 0.731$, $SD = 0.117$; $t(4343) = 4.54$, $p < .001$; Cohen's $d = 0.22$; Fig. 1). The mean difference was 0.025 (95% CI [0.016, 0.034]), corresponding to a 3.4% increase. The distributions overlapped substantially, indicating a shift in memorability rather than a categorically distinct high-memorability class. Both groups also spanned a broad range of ResMem scores (0.43–0.98), suggesting that mortality-related content elevates memorability on average without monopolizing the upper end of the scale.

Thus, mortality-related paintings showed a modest but reliable advantage in model-predicted memorability. Mortality-related status was determined from museum titles and curatorial descriptions using ChatGPT-assisted classification, followed by manual review of all selected mortality-related works. A supplementary robustness analysis used edge-based heuristics to detect paintings containing potentially text-like regions. Within mortality-related paintings, predicted memorability did not differ reliably between works flagged as containing such regions and those that were not. Because this procedure detected text-like edge patterns rather than verified text, it should be interpreted cautiously and only as a preliminary check that the main effect was unlikely to be driven solely by textual elements. We next asked whether this advantage was captured by hierarchical visual features, affective content, and variation across mortality-related subcategories.



Section 2. Mortality-related and non-death paintings are separable across the AlexNet hierarchy

We next asked whether mortality-related and non-death paintings were distinguishable in hierarchical visual representations. To test this, we extracted features from seven layers of an ImageNet-pretrained AlexNet (conv1-conv5, fc6, and fc7), reduced each layer to 100 principal components, and trained linear support vector machines to classify mortality-related versus non-death paintings using 5-fold cross-validation.

Mortality-related and non-death paintings were separable at every layer of the AlexNet hierarchy (Fig. 2). Classification accuracy was already high in early layers (conv1: 0.882; conv2: 0.894) and increased further in later layers, reaching a maximum at fc7 (0.915). This pattern indicates that mortality-related content is associated with stable visual structure that is detectable throughout the network but is most strongly represented in later layers. The progressive increase from early convolutional to later fully connected layers further suggests that the distinction is not limited to low-level image properties such as edges or textures but also involves higher-level visual organization.

To test whether this separation was also present in the geometry of the representation space itself, we conducted a matched-sample representational similarity analysis. This analysis yielded a convergent pattern: mortality-related and non-death paintings were increasingly separated in later layers, with the largest between-group separation observed at fc6 and fc7 (Fig. S1). Thus, the distinction was evident not only in supervised classification performance but also in the structure of the underlying feature space.

Together, these analyses indicate that hierarchical visual features capture substantial structure in mortality-related art. At the same time, separability alone does not establish that these features fully explain why mortality-related paintings show higher memorability. We therefore next tested how well AlexNet features predicted memorability within each group.

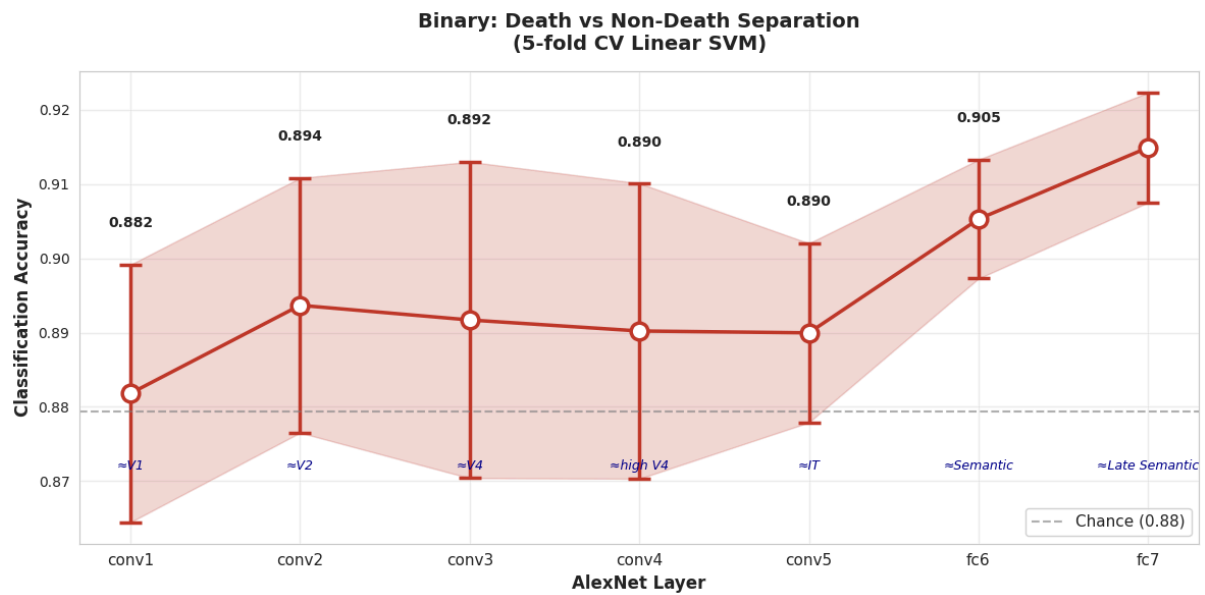


Figure 2. Death versus non-death separation across the AlexNet hierarchy.

PCA-reduced feature representations from each AlexNet layer were entered into 5-fold cross-validated linear support vector machine classifiers to distinguish mortality-related from non-death paintings. Classification accuracy was above the majority-class baseline at every layer and was highest in later layers, particularly fc6 and fc7, indicating robust separability across the network hierarchy.

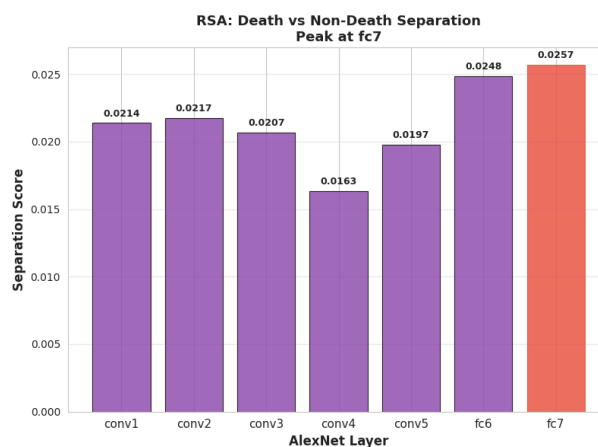


Figure S1. RSA-based separation of mortality-related and non-death paintings across AlexNet layers.

Separation scores were computed from matched-sample representational similarity analysis at each AlexNet layer. Higher values indicate greater between-group separation relative to within-group similarity. Mortality-related and non-death paintings were separable throughout the hierarchy, with the strongest separation in later layers, especially fc6 and fc7.

Section 3. Mortality-related memorability is less fully explained by hierarchical visual features

Having established that mortality-related and non-death paintings were separable across the AlexNet hierarchy, we next asked whether the same hierarchical visual features were

sufficient to explain variation in predicted memorability within each group. For each AlexNet layer, we used cross-validated ridge regression to predict ResMem scores from the PCA-reduced feature representations and quantified model performance as the correlation between predicted and observed memorability.

Across the full dataset, AlexNet features predicted memorability substantially, with performance increasing from early to later layers and reaching its maximum at fc7 ($r = 0.80$; Fig. 3). This pattern suggests that later network representations capture image properties that are especially relevant to memorability. The same general trend was evident within each group, but the strength of prediction differed markedly. For non-death paintings, prediction performance rose from $r = 0.53$ at conv1 to $r = 0.82$ at fc7. For mortality-related paintings, prediction also improved across the hierarchy but reached a lower maximum ($r = 0.62$ at the highest layer).

Thus, hierarchical visual features accounted for substantial variance in predicted memorability for both mortality-related and non-death paintings, but they accounted less fully for memorability in mortality-related works. Importantly, this result does not indicate that vision-based features fail for mortality-related art. Rather, it indicates that the memorability of mortality-related paintings is less tightly captured by the AlexNet feature hierarchy than is the memorability of non-death paintings. In other words, mortality-related paintings were more memorable on average (Section 1) and visually distinguishable across the network (Section 2), yet variation in their memorability was less fully explained by the same hierarchical features.

This asymmetry suggests that the memorability advantage of mortality-related paintings is not reducible to the visual structure captured by AlexNet alone. We therefore next asked whether affective content, particularly emotional arousal, could account for this residual advantage.

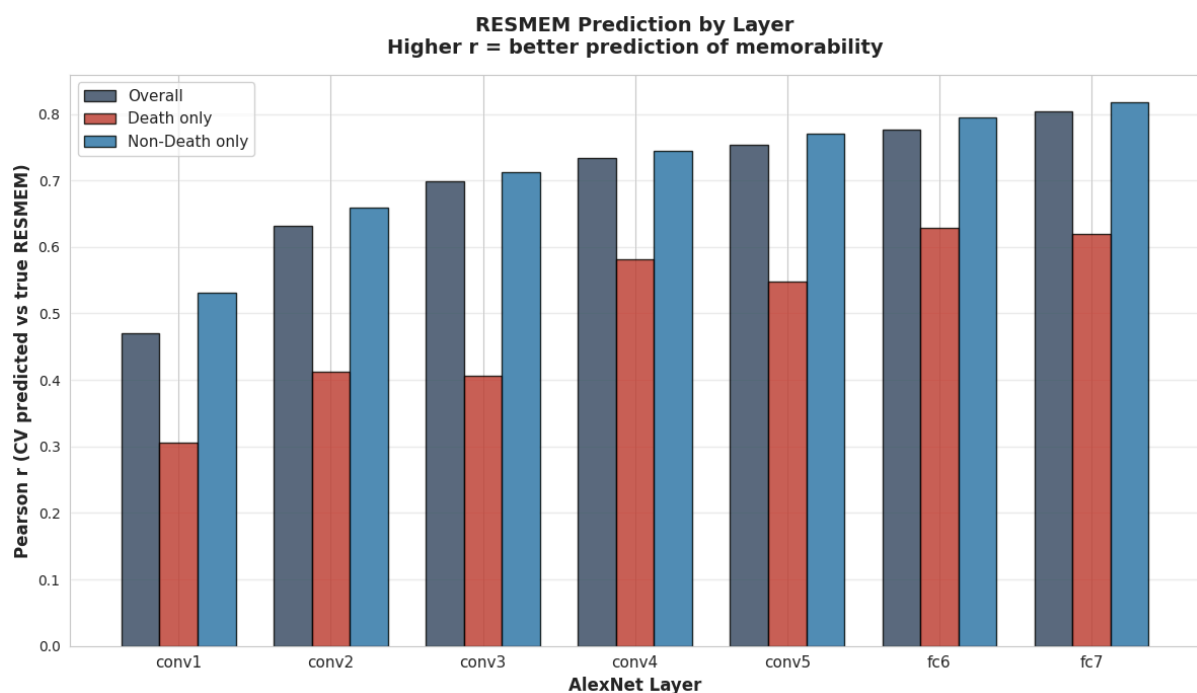


Figure 3. Prediction of ResMem scores from hierarchical AlexNet features.

For each AlexNet layer, PCA-reduced feature representations were used in 5-fold cross-validated ridge regression to predict ResMem scores, and performance was quantified as the Pearson correlation between predicted and observed memorability. Prediction performance increased progressively across the network for the full dataset, reaching its maximum at fc7. The same general pattern was observed within each group, but with a marked difference in magnitude: non-death paintings were predicted more strongly than mortality-related paintings at every layer, with the largest difference in the later fully connected layers (fc6-fc7). Thus, hierarchical visual features accounted for substantial variance in predicted memorability for both groups, but less fully for mortality-related than for non-death paintings.

Section 4. Emotional arousal mediates the mortality-related memorability advantage

The analyses above showed that mortality-related paintings were more memorable on average, visually distinguishable from non-death paintings across the AlexNet hierarchy, and less fully explained by those hierarchical visual features in terms of memorability prediction. We therefore next asked whether affective content could account for this memorability advantage. Given prior evidence linking emotional arousal to memory enhancement, we focused specifically on arousal as a candidate mediator of the association between mortality-related content and memorability.

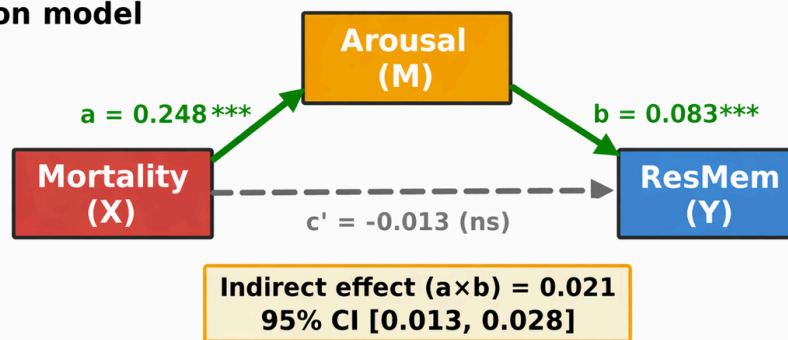
We tested a mediation model in which mortality-related status predicted memorability both directly and indirectly through arousal (Fig. 4). Mortality-related paintings were associated with higher arousal estimates (a path = 0.248, $p < .001$), and arousal in turn positively predicted memorability (b path = 0.083, $p < .001$). The indirect effect was significant ($ab = 0.021$, 95% CI [0.013, 0.028]). By contrast, the direct effect of mortality-related status on memorability was reduced to non-significance after arousal was included in the model ($c' = -0.013$, ns).

Thus, the memorability advantage of mortality-related paintings was statistically consistent with mediation by emotional arousal. In other words, mortality-related paintings were not more memorable simply by virtue of belonging to a semantic category labeled “death”; rather, their memorability advantage was closely tied to their higher affective arousal. This result helps explain why hierarchical visual features alone did not fully account for mortality-related memorability (Section 3): part of the advantage appears to be associated with affective salience rather than visual structure alone.

This mediation result also clarifies the interpretation of the mortality-related prediction gap. The gap is not evidence that vision-based models fail, but rather that they leave residual

variance associated with affective content.

a. Mediation model



b. Arousal is positively associated with memorability

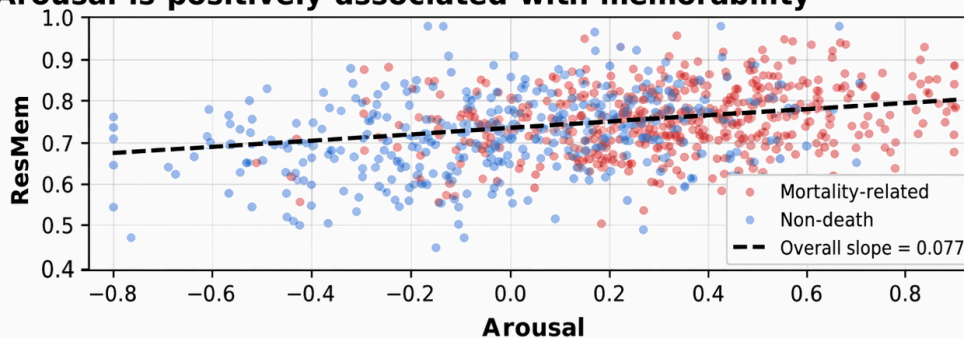


Figure 4. Emotional arousal mediates the mortality-related memorability advantage.

(A) Mediation model showing that mortality-related paintings were associated with higher arousal, which in turn predicted higher ResMem scores. The indirect effect was significant, whereas the direct effect of mortality-related status was reduced to non-significance after arousal was included. **(B)** Scatterplot showing the positive association between arousal and predicted memorability across paintings. Together, these results indicate that the memorability advantage of mortality-related paintings is statistically consistent with mediation by emotional arousal.

Section 5. Mortality-related subcategories show heterogeneous representational profiles across the AlexNet hierarchy

We next asked whether mortality-related paintings formed a single representational class or instead showed heterogeneity across subcategories. To test this, each mortality-related subcategory was classified against non-death paintings across AlexNet layers using PCA-reduced feature representations and cross-validated linear support vector machine classification.

The results revealed substantial heterogeneity across subcategories (Fig. 5A). Some mortality-related categories were highly separable from non-death paintings across much of the hierarchy. For example, the Hanno Series showed consistently strong separation at all layers and reached its highest value at fc7. Other/Unclear and Landscape also showed their strongest separation in later layers, whereas Religious and Death of [Person/Thing] peaked at FC6. By contrast, several categories showed peak separation earlier in the hierarchy: Funeral/Memorial, War/Violence, and Print Series (Other) peaked at conv5, whereas

Mythological/Historical Figures peaked at conv4. Symbolic/Personified Death was notable in showing particularly strong separation already at conv2, consistent with the possibility that some mortality-related imagery is distinguishable through relatively early or mid-level visual cues.

These results indicate that mortality-related content is not represented as a single visual-semantic class. Instead, different subcategories appear to be most distinguishable from non-death paintings at different stages of the AlexNet hierarchy. Some categories are most strongly differentiated in later layers, consistent with a larger contribution from higher-level visual or semantic organization, whereas others are already distinguishable in earlier or intermediate layers. This heterogeneity argues against a unitary account in which all mortality-related paintings share the same representational signature.

Heterogeneity was also evident when memorability was examined within subcategories. The features that best separated a mortality-related subcategory from non-death paintings were not always the same features that best predicted memorability within that subcategory (Fig. 5B). Thus, visual distinctiveness and memorability were related but not identical. Some subcategories could be strongly distinguished from non-death paintings while showing weaker memorability prediction, whereas others showed the opposite pattern. This dissociation reinforces the broader conclusion of Sections 3 and 4. That is, mortality-related memorability reflects identifiable visual structure but is not reducible to visual separability alone.

Together, these findings suggest that mortality-related paintings reach memorability through multiple representational routes rather than through a single shared pathway. Mortality-related content is therefore better understood as a heterogeneous family of image types that vary in how strongly they engage early visual, intermediate object-level, and later higher-level representations.

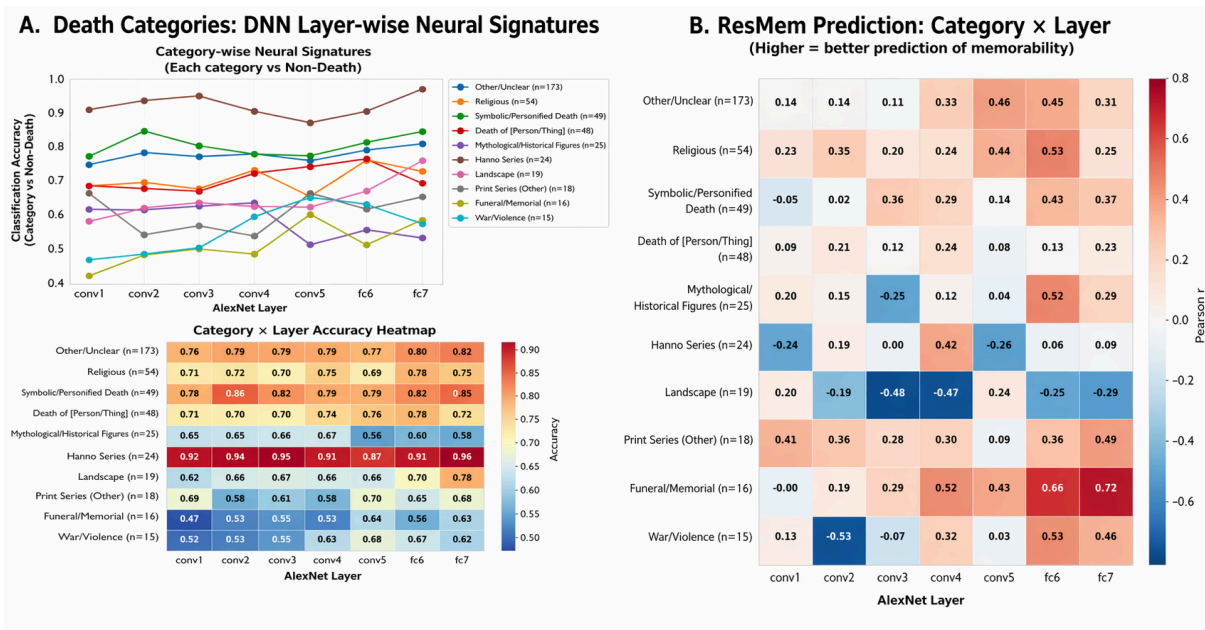


Figure 5. Mortality-related subcategories show heterogeneous profiles across AlexNet layers.

(A) Category-by-layer classification accuracy for each mortality-related subcategory versus non-death paintings.

(B) Category-by-layer prediction of ResMem within each mortality-related subcategory.

The two panels together indicate that mortality-related subcategories differ both in where they are most distinguishable from non-death paintings and in which layers best predict their memorability.

Discussion

The present study used mortality-related paintings as a test case for a general question: does hierarchical visual structure, as captured by current memorability models, fully account for image memorability, or does affective meaning contribute beyond it? Our findings argue for the latter. Across 4,345 paintings, death-themed works carried a modest but reliable memorability advantage, were visually distinguishable from non-death works throughout the AlexNet hierarchy, yet were less completely explained by those same hierarchical features than were non-death paintings. The residual gap was statistically consistent with mediation by emotional arousal. Together, these results suggest that predicted memorability reflects two partially distinct sources, identifiable visual structure and affective salience and that current vision-based models capture the former well while capturing the latter only indirectly.

Visual structure is substantial but insufficient

Two findings on visual structure are worth highlighting in combination. First, death-themed and non-death paintings were separable at every layer of AlexNet, with the strongest separation in later layers. This is consistent with the broader claim that memorability is embedded in image structure and can be predicted from hierarchical visual features (Khosla et al., 2015; Davis & Bainbridge, 2023). Second, and more informatively, the same features that distinguished the two groups predicted memorability less fully within death-themed works than within non-death works.

This asymmetry matters conceptually. The features that make an image recognizable as belonging to a class are not the same as the features that determine how memorable individual members of that class are. For non-death paintings, hierarchical visual features came close to exhausting what ResMem captures; for death-themed paintings, they did not. The latter, therefore, functions as a diagnostic case in which vision-based models show their limits, not by failing, but by leaving systematic residual variance that other content cues help explain.

A dual contribution from visual and affective content

The mediation analysis offers a candidate explanation for this residual variance. Death-themed paintings were associated with higher estimated arousal, arousal in turn predicted higher memorability, and the direct effect of mortality-related status was reduced to non-significance once arousal was included. Although mediation with cross-sectional, model-derived variables cannot establish causality, the pattern is consistent with the possibility that what makes mortality-related art memorable is not just that it can be visually recognized as such, but that it is affectively consequential. This interpretation aligns with long-standing evidence that emotionally arousing material receives prioritized encoding and

consolidation (LaBar & Cabeza, 2006; McGaugh, 2015), and it offers a concrete mechanism for the gap observed in Section 3: arousal is encoded only weakly, if at all, in an object-recognition network pretrained on ImageNet.

The resulting picture is a dual-contribution account. Death-themed art carries visual-semantic structure that current models capture, and it additionally carries affective salience that those models do not directly represent but that contributes measurable variance in predicted memorability. Importantly, this is not a claim about death per se; it is a claim about how affectively potent content interacts with the representational assumptions of vision-based memorability models. Other emotionally salient categories, i.e., erotic, threatening, or morally charged imagery, might show an analogous pattern, and this is a testable direction.

Mortality-related content is a heterogeneous family

The subcategory analyses qualify the interpretation in an important way. Death-themed works did not behave as a single representational class. Subcategories reached peak separation from non-death paintings at different stages of the hierarchy, i.e.g, Symbolic/Personified Death already at conv2, Mythological/Historical Figures at conv4, Funeral/Memorial and War/Violence at conv5, and Religious and the Hanno Series at the later fully connected layers. Moreover, the layer that best separated a subcategory from non-death works was not always the layer that best predicted memorability within it.

This dissociation has two implications. First, it argues against a single "death signature" in hierarchical vision: different kinds of death-related imagery depend on different mixtures of low-, mid-, and high-level visual cues. Skeletal or allegorical symbols may be detectable from early visual features because they contain distinctive shape primitives; narrative scenes of violence or mourning likely require object- and scene-level integration; religious depictions and thematically unified series engage higher-level semantic organization. Second, visual distinctiveness and memorability are related but dissociable. A subcategory can be visually atypical without being especially memorable, or memorable without being maximally separable, reinforcing the broader point that current models index what images look like more fully than why they stick.

Implications for memorability models and emotional memory

Two broader implications follow. For computational models of memorability, our findings suggest that performance can plateau in systematic rather than random ways. Where residual variance exists, it may not be noise; it may index image properties, most notably affective salience, which a purely vision-based model does not encode directly. One promising direction is to augment stimulus-based models with content signals derived from vision-language representations, which already align images with affective or semantic descriptors and might therefore recover part of the variance left unexplained here. Another is to evaluate whether newer architectures (e.g., CLIP, vision transformers, or multimodal models) close the mortality-related gap; the present findings make a specific, falsifiable prediction that they should do so to the extent that they encode affective content.

For research on emotional memory, our results show that the well-established arousal-memory link is detectable even at the level of stimulus-intrinsic, model-predicted memorability, without direct behavioral measurement. This is theoretically interesting because it locates part of the effect in the stimulus itself, rather than solely in encoding or retrieval processes. It also provides a natural bridge between the memorability tradition, which has emphasized properties of images, and the emotional-memory tradition, which has emphasized properties of encoding. Our results suggest these two traditions are converging on overlapping terrain.

We treat the link to terror management theory cautiously. The present design cannot test whether death-related content is psychologically privileged in the specific sense TMT proposes; it can only show that death-themed paintings, as a class, are more memorable than other paintings and that this advantage is tied to affective arousal. Whether mortality-related meaning is distinctive beyond generic emotional arousal, for example, by engaging self-relevant or existential processing, is a question for behavioral and cross-cultural work, not for a computational analysis of painting metadata.

Limitations and next steps

Three limitations are important. First, our outcome measure is model-predicted memorability, not behavioral recognition memory. ResMem has been validated against human performance for artworks in both online and in-person settings (Davis & Bainbridge, 2023), but the present findings are computational evidence about memorability rather than direct evidence from memory experiments. Second, arousal was estimated via CLIP prompts rather than human ratings, and the mediation analysis therefore involves two model-derived constructs; convergent evidence with behavioral arousal ratings is needed. Third, mortality-related status was determined using a language-model-assisted classification procedure with manual review. This is practical at the scale of thousands of paintings but inherits whatever biases exist in museum titles and descriptions, and some subcategories had small sample sizes that limit the stability of category-level estimates.

These limitations motivate specific next steps. We are currently preparing a cross-cultural behavioral experiment in which participants in China and Canada view a balanced set of death-themed and non-death paintings and perform continuous recognition, rate arousal and valence, and complete individual-difference measures relevant to mortality salience. This design will (i) test whether the memorability advantage reported here replicates behaviorally, (ii) validate the CLIP-derived arousal estimates against human ratings, and (iii) examine whether the effect and its mediation by arousal generalize across cultural contexts in which death is culturally foregrounded or culturally avoided. In parallel, the computational analyses reported here can be extended to newer vision and vision-language architectures, providing a direct test of whether the prediction gap narrows as models incorporate more affective or semantic content.

Conclusion

Memorability research has made substantial progress by treating memorability as a stimulus property that can be predicted from images. The present study suggests that this progress is real but bounded: for affectively charged content, part of what makes an image memorable

lies in meaning that current vision-based models encode only incompletely. Using mortality-related paintings as a test case, we find that visual structure explains much of the memorability signal, but that affective arousal accounts for the portion that remains. Memorability may therefore be best understood as the joint product of what an image looks like and what it means to see it.

Methods

Dataset and sample construction

We analyzed paintings from the Art Institute of Chicago (AIC). The non-death comparison set was drawn from the AIC painting subset used in prior memorability research on artworks, in which more than 4,000 AIC paintings were assigned memorability estimates and shown to have reliable memory performance across online and in-person settings. ResMem, the memorability model used in that work, has been validated against human memory performance for artworks and is described by the Bainbridge lab as a state-of-the-art model for estimating intrinsic image memorability (Davis and Bainbridge, 2023).

To increase coverage of mortality-related content, we separately screened the broader AIC painting collection for mortality-related works. The final analytic sample contained 4,345 paintings, including 488 mortality-related paintings and 3,857 non-death paintings. All analyses were restricted to paintings with available image files and memorability scores.

Mortality-related classification

Mortality-related status was determined from museum-provided metadata, including titles and curatorial descriptions. Candidate mortality-related works were identified from titles and descriptions containing mortality-relevant themes and terms (e.g., *death*, *funeral*, *crucifixion*), using a ChatGPT-assisted classification procedure informed by both textual annotations and the images themselves. All selected mortality-related paintings were then manually reviewed by the author to confirm inclusion and refine category assignments.

In addition to the binary mortality-related versus non-death distinction, mortality-related paintings were assigned to subcategories for category-level analyses. These included Other/Unclear, Religious, Symbolic/Personified Death, Death of [Person/Thing], Mythological/Historical Figures, Hanno Series, Landscape, Print Series (Other), Funeral/Memorial, and War/Violence.

Predicted memorability

Predicted memorability was quantified using ResMem. Prior memorability work has shown that image memorability is a stable, stimulus-driven property of images, and later work has demonstrated that ResMem predicts human memory performance for artworks in both online experiments and naturalistic museum viewing. Accordingly, ResMem scores were treated here as model-predicted memorability, not as direct behavioral memory performance, but as a behaviorally meaningful proxy grounded in prior validation work (Davis & Bainbridge, 2023).

Affective content estimates

Affective content was estimated using CLIP (openai/clip-vit-base-patch32), a vision-language model trained to align images and text in a shared embedding space (Radford et al., 2021). For each painting, we used a zero-shot prompt-based procedure to derive image-level emotion probabilities, from which valence and arousal scores were computed. Valence was defined as the balance of positive versus negative affective prompt probabilities, and arousal as the balance of high- versus low-arousal prompt probabilities. Additional implementation details, including the specific prompts used, are reported in the code folder.

Image preprocessing and AlexNet feature extraction

To characterize hierarchical visual structure, we extracted features from an ImageNet-pretrained AlexNet. AlexNet is a deep convolutional neural network with five convolutional layers and three fully connected layers, originally trained on ImageNet and widely used as a canonical hierarchical vision model (Krizhevsky et al., 2015).

Images were resized to 256 pixels, center-cropped to 224×224 pixels, converted to RGB, transformed to tensors, and normalized using standard ImageNet channel means and standard deviations. Layer-wise features were extracted from seven AlexNet stages: conv1, conv2, conv3, conv4, conv5, fc6, and fc7. For each image, features were flattened into one-dimensional vectors.

Because these representations were high-dimensional, each layer was reduced separately using principal components analysis (PCA), retaining up to 100 principal components per layer. These PCA-reduced feature sets were used in all subsequent DNN analyses.

Binary classification across the AlexNet hierarchy

To test whether mortality-related and non-death paintings were distinguishable in hierarchical visual representations, we trained linear support vector machine (SVM) classifiers separately at each AlexNet layer. For each layer, PCA-reduced features were z-scored and entered into 5-fold cross-validated classification. Performance was quantified as mean accuracy across folds. Because the classes were imbalanced, we also reported the majority-class baseline for reference.

Representational similarity analysis

To examine whether mortality-related and non-death paintings were also separated in the geometry of the representation space itself, we conducted a representational similarity analysis (RSA). Mortality-related paintings were matched with an equal number of randomly sampled non-death paintings. Within each AlexNet layer, pairwise correlation distances were computed, yielding a representational distance matrix. Mean within-group and between-group distances were then summarized with a normalized separation index:

$$\text{separation} = \frac{\text{between} - \text{within}_{avg}}{\text{between} + \text{within}_{avg}}$$

where within_{avg} is the average of the mortality-related and non-death within-group distances. Higher values indicate greater between-group separation relative to within-group similarity.

Prediction of memorability from AlexNet features

To test whether hierarchical visual features explained memorability similarly for mortality-related and non-death paintings, we predicted ResMem scores from PCA-reduced AlexNet features using ridge regression. For each AlexNet layer, features were standardized and entered into 5-fold cross-validated ridge regression. Prediction performance was quantified as the Pearson correlation between predicted and observed memorability.

This analysis was conducted three ways: across the full dataset, within the mortality-related subset only, and within the non-death subset only. The resulting layer-wise prediction profiles were compared to evaluate whether visual features accounted for memorability equally well across groups.

Mediation analysis

We tested whether emotional arousal statistically mediated the association between mortality-related status and predicted memorability using ordinary least squares regression in statistical models. We estimated:

- (1) the total effect of mortality-related status on memorability;
- (2) the effect of mortality-related status on arousal; and
- (3) the joint model predicting memorability from mortality-related status and arousal, yielding the mediator effect and the direct effect controlling for arousal.

To estimate its uncertainty, we used nonparametric bootstrap resampling with 5,000 resamples and calculated the 95% percentile confidence interval.

Category-level analyses

To examine heterogeneity across mortality-related content, we repeated the AlexNet analyses at the subcategory level. Only categories with at least 15 paintings were retained for the main category-level analyses. For each eligible subcategory, linear SVM classifiers were trained to distinguish that subcategory from non-death paintings across AlexNet layers. Because subcategory sizes were much smaller than the non-death set, non-death paintings were subsampled for these comparisons, and the number of folds was reduced when necessary.

We also examined memorability prediction within each mortality-related subcategory. For each category and layer, PCA-reduced AlexNet features were used in cross-validated ridge

regression to predict ResMem scores, and performance was quantified as the Pearson correlation between predicted and observed memorability.

References

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